

3.2.1 Enthalpy changes

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
Enthalpy changes: ΔH of reaction, formation, combustion and neutralisation	
(a) explanation that some chemical reactions are accompanied by enthalpy changes that are exothermic (ΔH , negative) or endothermic (ΔH , positive)	
(b) construction of enthalpy profile diagrams to show the difference in the enthalpy of reactants compared with products	M3.1

- (c) qualitative explanation of the term *activation energy*, including use of enthalpy profile diagrams
- (d) explanation and use of the terms:
- (i) *standard conditions* and *standard states* (physical states under standard conditions)
 - (ii) *enthalpy change of reaction* (enthalpy change associated with a stated equation, $\Delta_r H$)
 - (iii) *enthalpy change of formation* (formation of 1 mol of a compound from its elements, $\Delta_f H$)
 - (iv) *enthalpy change of combustion* (complete combustion of 1 mol of a substance, $\Delta_c H$)
 - (v) *enthalpy change of neutralisation* (formation of 1 mol of water from neutralisation, $\Delta_{\text{neut}} H$)
- (e) determination of enthalpy changes directly from appropriate experimental results, including use of the relationship: $q = mc\Delta T$

Bond enthalpies

- (f) (i) explanation of the term *average bond enthalpy* (as the breaking of 1 mol of bonds in gaseous molecules)
- (ii) explanation of exothermic and endothermic reactions in terms of enthalpy changes associated with the breaking and making of chemical bonds
- (iii) use of average bond enthalpies to calculate enthalpy changes and related quantities (see also 2.2.2 f)

Hess' law and enthalpy cycles

- (g) Hess' law for construction of enthalpy cycles and calculations to determine indirectly:
- (i) an enthalpy change of reaction from enthalpy changes of combustion
 - (ii) an enthalpy change of reaction from enthalpy changes of formation
 - (iii) enthalpy changes from unfamiliar enthalpy cycles
- (h) the techniques and procedures used to determine enthalpy changes directly and indirectly.

M3.1

Activation energy in terms of the minimum energy required for a reaction to take place.

Definitions required for enthalpy changes of formation, combustion and neutralisation only.

Standard conditions can be considered as 100 kPa and a stated temperature, 298 K.

M0.0, M0.2, M2.2, M2.3, M2.4

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M0.0, M0.2, M2.2, M2.3, M2.4

Formal definition of average bond enthalpy **not** required.

Learners are expected to understand that an actual bond enthalpy may differ from the average value.

M0.0, M0.2, M1.1, M2.2, M2.3, M2.4, M3.1

Definition of Hess' law **not** required. Unfamiliar enthalpy cycles will be provided.

HSW2 Application of the principle of conservation of energy to determine enthalpy changes.

M3.1, M3.2

To include the enthalpy changes covered in 5.2.1 c.

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HSW4 Opportunities for carrying out experimental and investigative work.